Python Modules

Hands-on Experience — Standard Library

1. Importing and Exploring a Module

The **import** keyword loads a module so its functions and variables become available.

Example

```
import math

print(math.sqrt(9))
print(math.pi)
```

Output

```
3.0
3.141592653589793
```

Use **help()** to read documentation for a module or a specific function, and **dir()** to list everything a module exposes.

Example — help & dir

```
import math

help(math)          # full module docs
help(math.sqrt)     # docs for one function
print(dir(math))    # list all names in math
```

2. Importing a Specific Function

Use **from … import** to pull in only what you need. The function can then be called without the module prefix.

Example — sqrt

```
from math import sqrt

print(sqrt(9))
```

```
3.0
```

Example — factorial

```
from math import factorial

print(factorial(5))
```

```
120
```

3. Renaming a Module Using as

The **as** keyword creates an alias for a module or a function.

Alias for a module

```
import math as maths

print(maths.sqrt(9))
print(maths.pi)
```

```
3.0
3.141592653589793
```

Alias for a function

```
from math import sqrt as squareroot

print(squareroot(9))
```

```
3.0
```

■ *import math.sqrt is invalid — sqrt is a function, not a sub-module.*

4. Time Module

The **time** module provides functions for querying and formatting the current time, and for pausing execution.

time.time() — seconds since Unix epoch (Jan 1 1970)

```
import time

current_time = time.time()
print("Seconds since epoch:", current_time)
```

time.ctime() — human-readable timestamp

```
import time

print(time.ctime())          # e.g.  Tue Sep  8 10:30:00 2026
```

time.sleep() — pause execution

```
import time

print("Program started")
time.sleep(3)            # pause for 3 seconds
print("3 seconds have passed")

# Decimal values are allowed
time.sleep(0.5)          # pause for half a second
```

5. Glob Module

The **glob** module finds files and folders matching a pattern. The wildcard ***** matches any number of characters.

Common patterns

```
import glob

print(glob.glob("*"))        # all files and folders
print(glob.glob("*.py"))     # all Python files
print(glob.glob("*.txt"))    # all text files
print(glob.glob("test*"))    # files starting with 'test'
```

***.py** would match: *program.py*, *hello.py*, *lab01.py* etc.

6. Random Module

The **random** module generates random numbers and performs random selections.

random.randint(a, b) — random integer, inclusive of both ends

```
import random

print(random.randint(1, 10))   # e.g. 7
```

random.random() — float in [0, 1)

```
import random

print(random.random())   # e.g. 0.736421
```

random.shuffle(list) — rearrange a list in-place

```
import random

x = [1, 2, 3, 4, 5]
random.shuffle(x)
print(x)   # e.g. [3, 5, 1, 4, 2]
```

random.sample(list, k) — pick k unique elements (original unchanged)

```
import random

x = [1, 2, 3, 4, 5]
result = random.sample(x, 2)

print("Original list:    ", x)
print("Selected elements:", result)
```

```
Original list:     [1, 2, 3, 4, 5]
Selected elements: [2, 5]
```

Summary: import, from … import, as · dir() / help() · math · time · glob · random